EE 3EY4
Lab 6

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Activity 1:

Format: $x \quad y \quad z \quad q_x \quad q_y \quad q_z \quad q_w$

$$T^{base_link}_{imu} \begin{cases} t^A_B = \langle 0.01286 \quad 0 \quad 0.055 + 0.00614 = 0.06119 \rangle \\ \hat{u} = \langle 0 \quad 0 \quad 1 \rangle \\ \alpha^A_B = +90^\circ = +\frac{\pi}{2} \text{ rad} \end{cases}$$

$$q = \sin\left(\frac{\alpha}{2}\right)(xi + yj + zk) + \cos\left(\frac{\alpha}{2}\right)$$
$$= \sin\left(\frac{\pi}{4}\right)(0i + 0j + 1k) + \cos\left(\frac{\pi}{4}\right)$$
$$= 0i + 0j + 0.707k + 0.707$$
$$q = q_x i + q_y j + q_z k + q_w$$

$$T^{base_link}_{laser} \begin{cases} t^A_B = \langle 0.01286 \quad 0 \quad 0.055 + 0.021 = 0.076 \rangle \\ \hat{u} = \langle 0 \quad 0 \quad 1 \rangle \\ \alpha^A_B = +180^\circ = +\pi \text{ rad} \end{cases}$$

$$q = \sin\left(\frac{\alpha}{2}\right)(xi + yj + zk) + \cos\left(\frac{\alpha}{2}\right)$$
$$= \sin\left(\frac{\pi}{2}\right)(0i + 0j + 1k) + \cos\left(\frac{\pi}{2}\right)$$
$$= 0i + 0j + 1k + 0$$
$$q = q_x i + q_y j + q_z k + q_w$$

$$T^{base_link}_{camera} \begin{cases} t^A_B = \langle 0.27017 + 0.009 + 0.01286 = 0.29203 \quad 0 \quad 0.055 + 0.031 = 0.086 \rangle \\ \hat{u} = \\ \alpha^A_B = \end{cases} \Big\} q$$

$$R^{base_link}_{camera} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\alpha^A_B = \cos^{-1}\left(\frac{r_{11} + r_{22} + r_{33} - 1}{2}\right) = \cos^{-1}\left(\frac{0 + 0 + 0 - 1}{2}\right) = 120^\circ = \frac{2\pi}{3} \text{ rad or } -120^\circ = -\frac{2\pi}{3} \text{ rad}$$

$$\hat{u} = \frac{1}{2\sin(\alpha)} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \frac{1}{2\sin\left(\frac{2\pi}{3}\right)} \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \Rightarrow xi + yj + zk$$

$$R^{base_link}_{camera} \longrightarrow q$$

$$q = \sin\left(\frac{\alpha}{2}\right)(xi + yj + zk) + \cos\left(\frac{\alpha}{2}\right)$$

$q = -0.5i + 0.5j - 0.5k + 0.5$ or $q = 0.5i - 0.5j + 0.5k - 0.5$

Both same

```
# Complete the missing parts in the homogenous transformations below (Activity #1)

 <node pkg="tf2_ros" type="static_transform_publisher" name="base_link_to_imu"
        args="0.01286 0.0 0.06119 0.0 0.0 0.707 0.707 base_link imu" />

 <node pkg="tf2_ros" type="static_transform_publisher" name="base_link_to_laser"
        args="0.01286 0.0 0.076 0.0 0.0 1.0 0.0 base_link laser" />

<node pkg="tf2_ros" type="static_transform_publisher" name="base_link_to_camera"
        args="0.29203 0.0 0.086 0.5 -0.5 0.5 -0.5 base_link camera_link" />
```

Activity 2:

Quaternions serve as a mathematically compact method for encoding the rotation between a reference frame and a target frame. In this activity, the base_link serves as the primary reference, with its x-axis oriented forward and its z-axis oriented upward. Because peripheral sensors such as the IMU, LiDAR, and camera are mounted in different orientations, their local axes do not align with the vehicle's coordinate system. To fully define these relationships, we utilize Homogeneous Transformations, which consist of both a translation component and a rotation component. The translation defines the x, y, and z distances from the base_link origin to the origin of the target sensor frame, while the rotation is defined by a four-element quaternion vector that defines the angular orientation of the sensor relative to the vehicle.

Using quaternions over traditional Euler angles offers several critical benefits for real-time robotics. One of the primary advantages is the avoidance of Gimbal Lock, a state where two rotational axes align and cause the loss of one degree of freedom, which quaternions bypass by representing orientation in four-dimensional space. Additionally, quaternions support Spherical Linear Interpolation (Slerp), which is essential for maintaining smooth motion and accurate sensor fusion when dealing with continuous vehicle movement. Finally, they are more computationally efficient and numerically stable for frequent rotation compositions and normalizations, making them ideal for the high-frequency processing required by the MacAEV's autonomous systems.

Activity 4:

.bag file uploaded on Avenue

Activity 5:

experiment.cpp file uploaded on Avenue

Activity 6:

Screenshots of RVIZ Window:

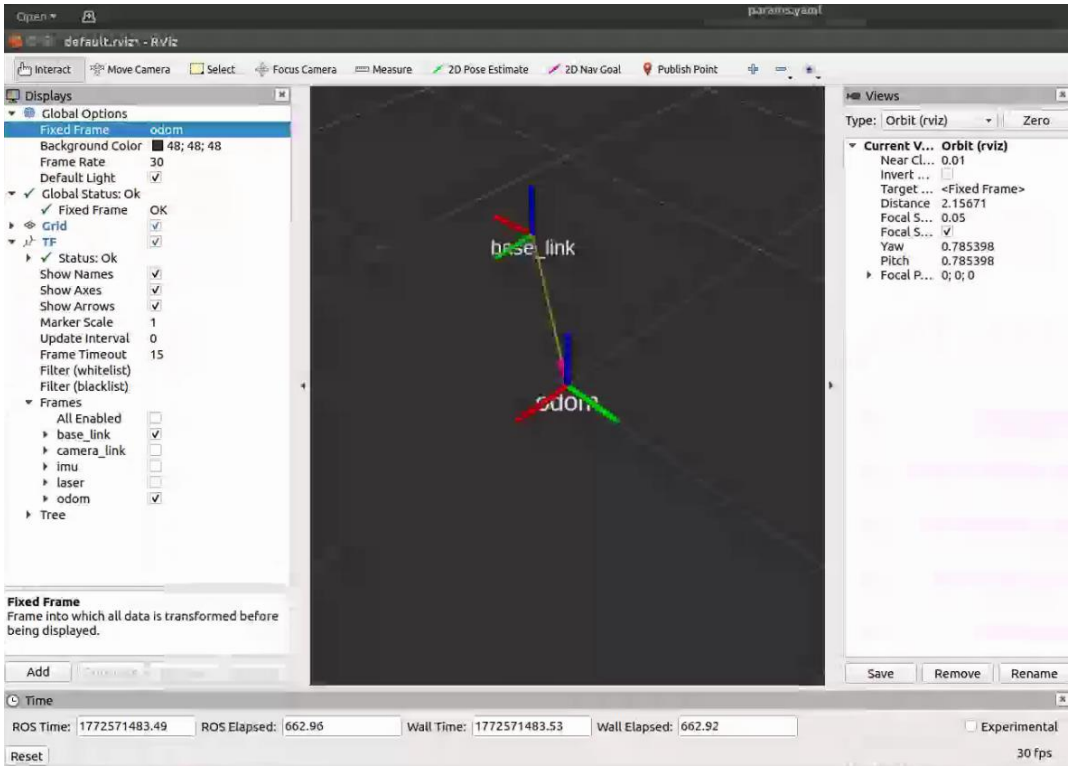


Figure 1: Driving from initial point

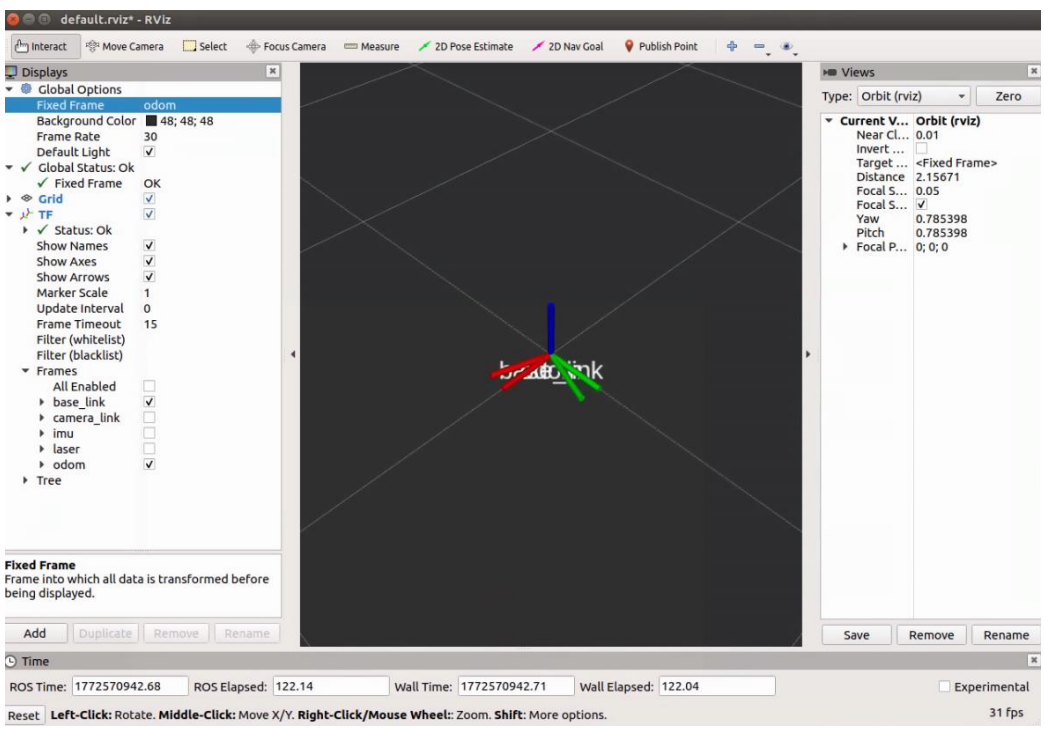


Figure 2: Returning to original point

To evaluate the performance of our system, we marked the vehicle's starting point, drove a test circuit (In the path of a hemisphere), and attempted to return to the exact same physical coordinates. As observed in Figure 2, the odom and base_link frames were not perfectly aligned upon return, indicating a measurable pose error. This discrepancy primarily stems from our system calculating the vehicle's position by integrating velocity and steering commands over time.

Any extremely small error in these calculations or the time-step measurements accumulates over the duration of the experiment. Furthermore, wheel odometry is highly vulnerable to physical disturbances such as wheel slip on smooth surfaces or momentary loss of traction on uneven terrain. Since the system assumes every wheel rotation corresponds to a specific linear distance, these physical inconsistencies lead to "drift," where the estimated pose increasingly diverges from the vehicle's actual position as the total travel distance grows.

.bag file uploaded on Avenue (re-recorded as original .bag file was empty)

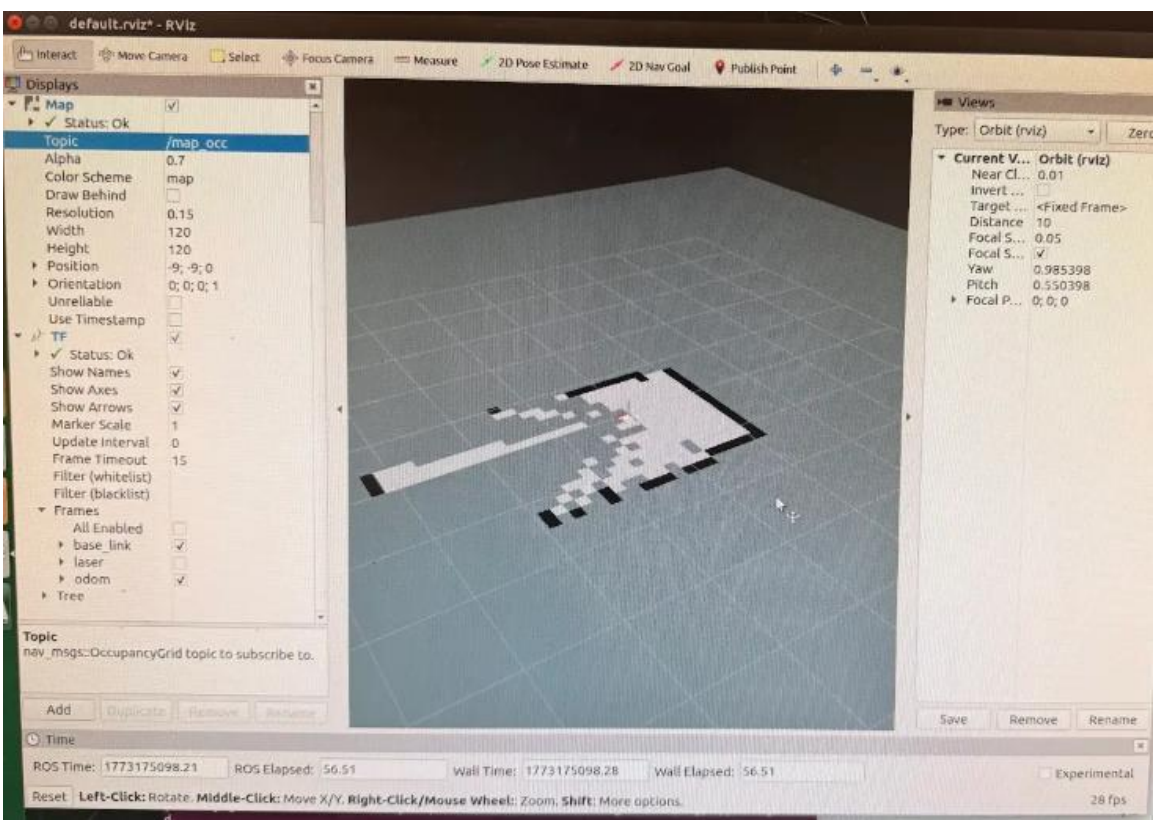
Activity 7:

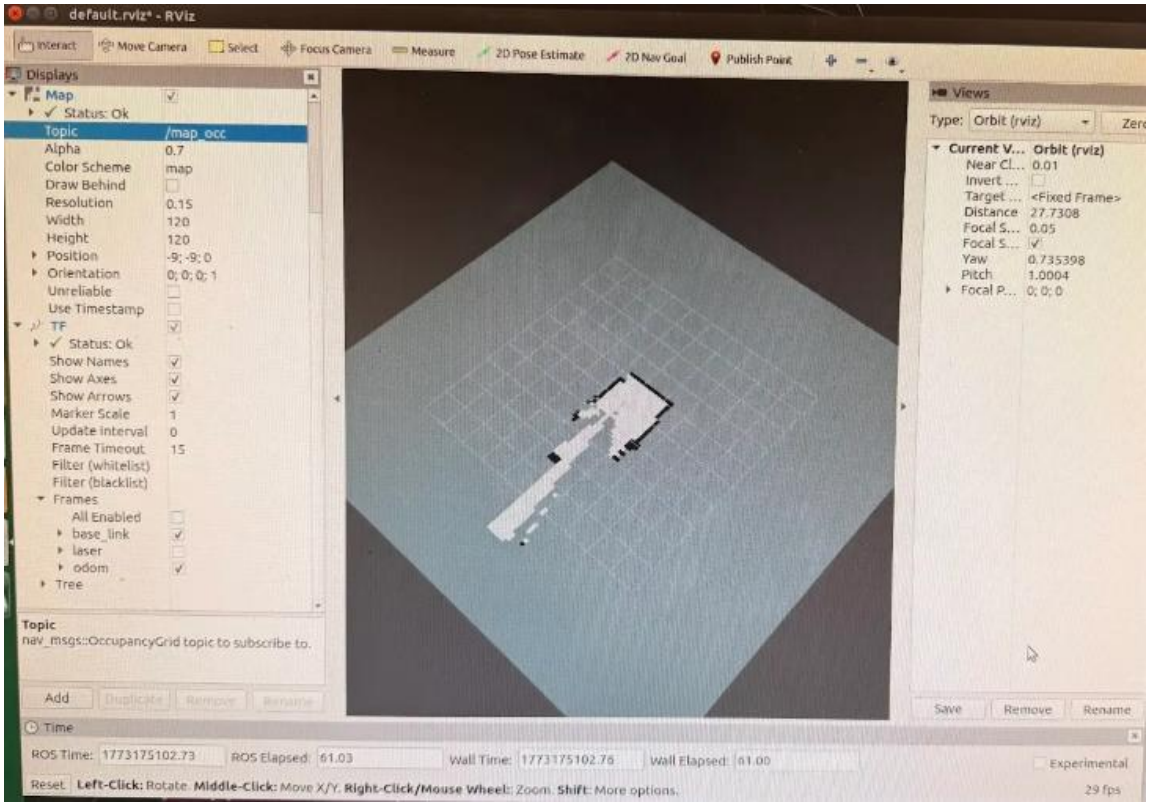
Occupancygridmap.py file uploaded on Avenue

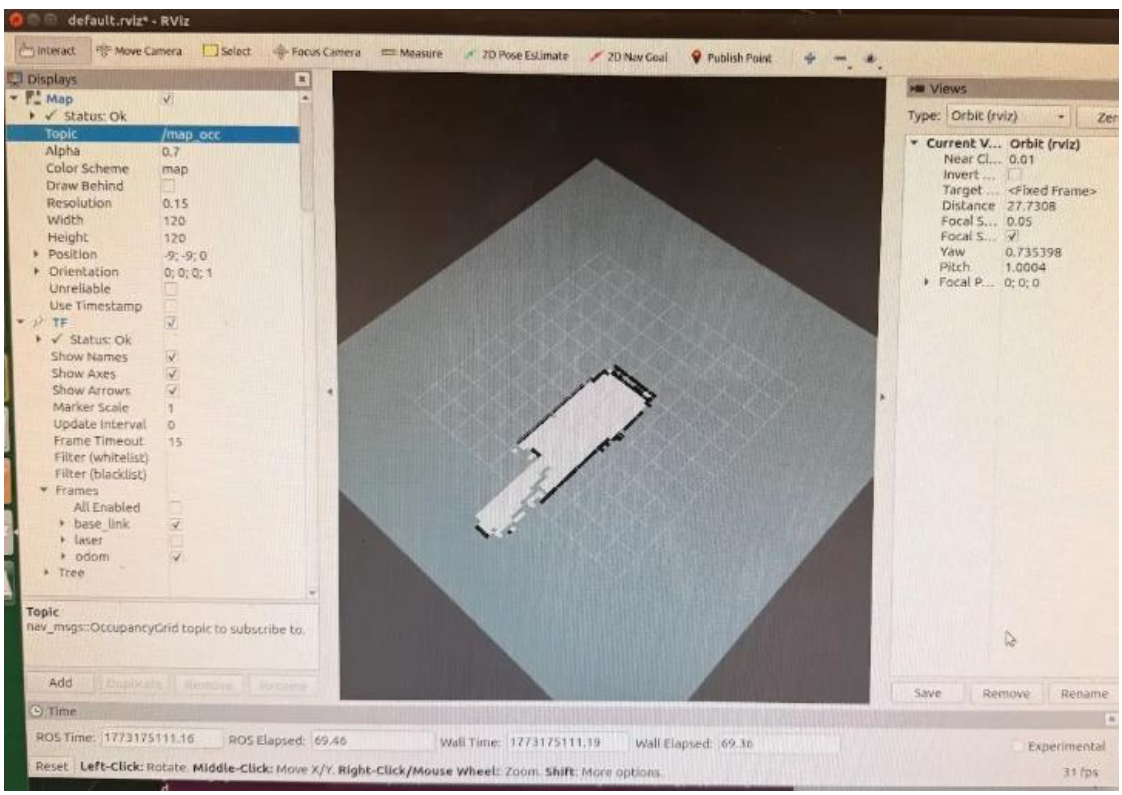
Activity 8:

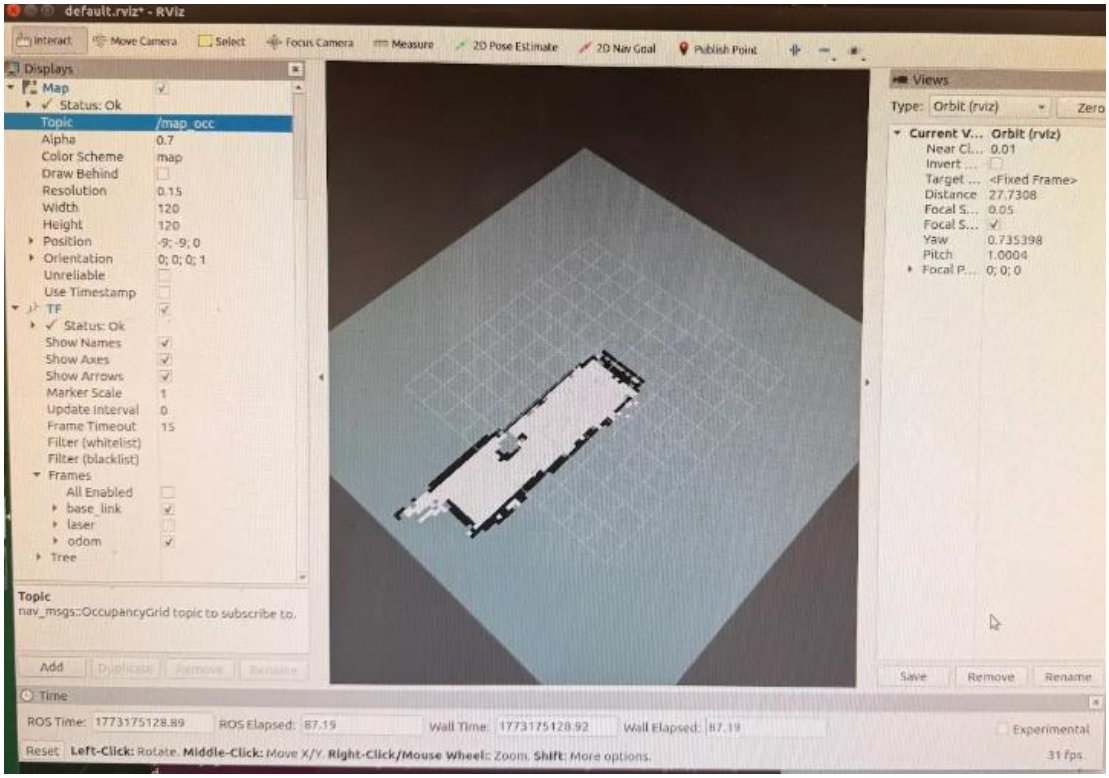
.bag file uploaded on Avenue

Images of our results in sequence:




default.rviz* - RViz
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Nav Goal
Publish Point
Displays
Map
Status: Ok
Topic /map_occ
Alpha 0.7
Color Scheme map
Draw Behind
Resolution 0.15
Width 120
Height 120
Position -9; -9; 0
Orientation 0; 0; 0; 1
Unreliable
Use Timestamp
TF
Status: Ok
Show Names
Show Axes
Show Arrows
Marker Scale 1
Update Interval 0
Frame Timeout 15
Filter (whitelist)
Filter (blacklist)
Frames
All Enabled
base_link
laser
odom
Tree
Topic
nav_msgs::OccupancyGrid topic to subscribe to.
Add
Duplicate
Remove
Rename
Views
Type: Orbit (rviz)
Current V... Orbit (rviz)
Near Cl... 0.01
Invert ...
Target ... <Fixed Frame>
Distance 27.7308
Focal S... 0.05
Focal S...
Yaw 0.735398
Pitch 1.0004
Focal P... 0; 0; 0
Save
Remove
Rename
Time
ROS Time: 1773175102.73
ROS Elapsed: 61.03
Wall Time: 1773175102.76
Wall Elapsed: 61.00
Experimental
Reset
Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click/Mouse Wheel:: Zoom. Shift: More options.
29 fps


default.rviz* - RViz
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Nav Goal
Publish Point
Displays
Map
Status: Ok
Topic /map_occ
Alpha 0.7
Color Scheme map
Draw Behind
Resolution 0.15
Width 120
Height 120
Position -9; -9; 0
Orientation 0; 0; 0; 1
Unreliable
Use Timestamp
TF
Status: Ok
Show Names
Show Axes
Show Arrows
Marker Scale 1
Update Interval 0
Frame Timeout 15
Filter (whitelist)
Filter (blacklist)
Frames
All Enabled
base_link
laser
odom
Tree
Topic
nav_msgs::OccupancyGrid topic to subscribe to.
Add
Duplicate
Remove
Rename
Views
Type: Orbit (rviz)
Current V... Orbit (rviz)
Near Cl... 0.01
Invert ...
Target ... <Fixed Frame>
Distance 27.7308
Focal S... 0.05
Focal S...
Yaw 0.735398
Pitch 1.0004
Focal P... 0; 0; 0
Save
Remove
Rename
Time
ROS Time: 1773175111.16
ROS Elapsed: 69.46
Wall Time: 1773175111.19
Wall Elapsed: 69.36
Experimental
Reset
Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click/Mouse Wheel:: Zoom. Shift: More options.
31 fps

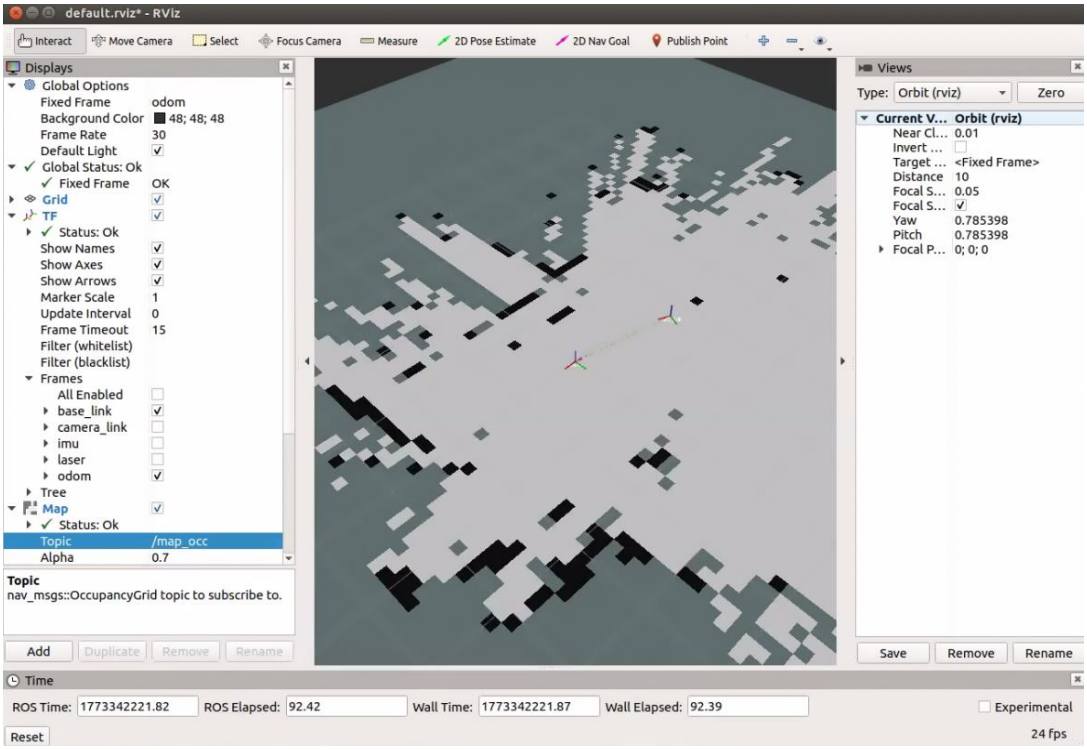


Activity 9:

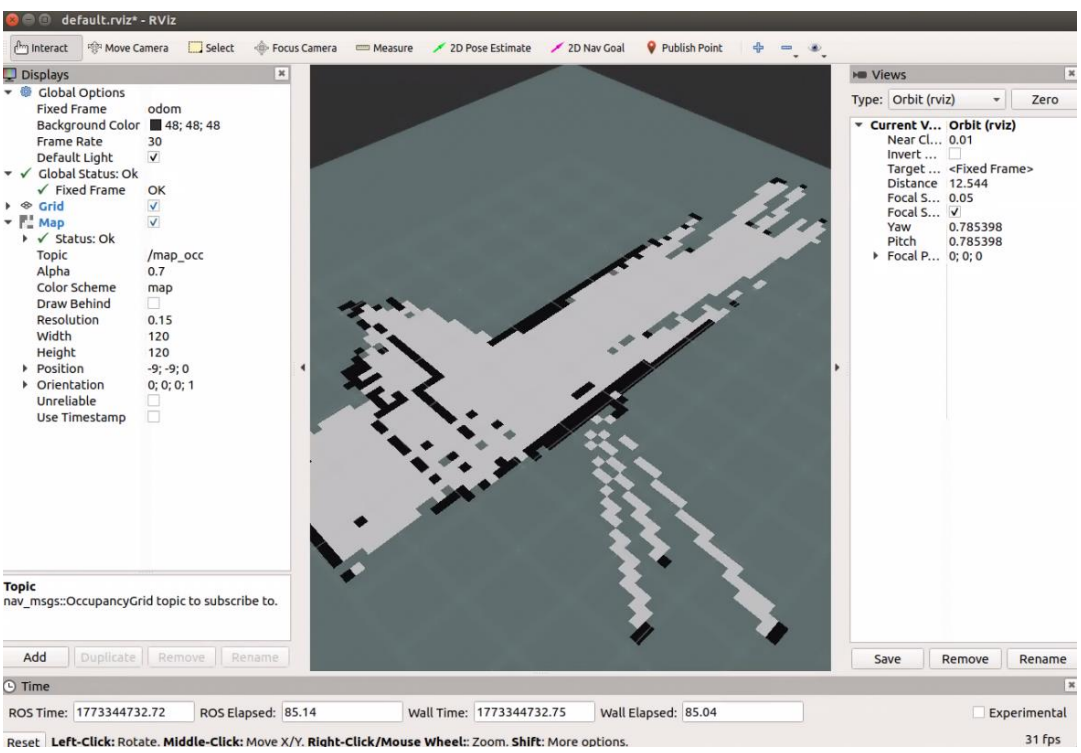
.bag file uploaded on Avenue. (Had to be re-recorded as original .bag file was empty)

Hallway reference of what the vehicle was mapping:

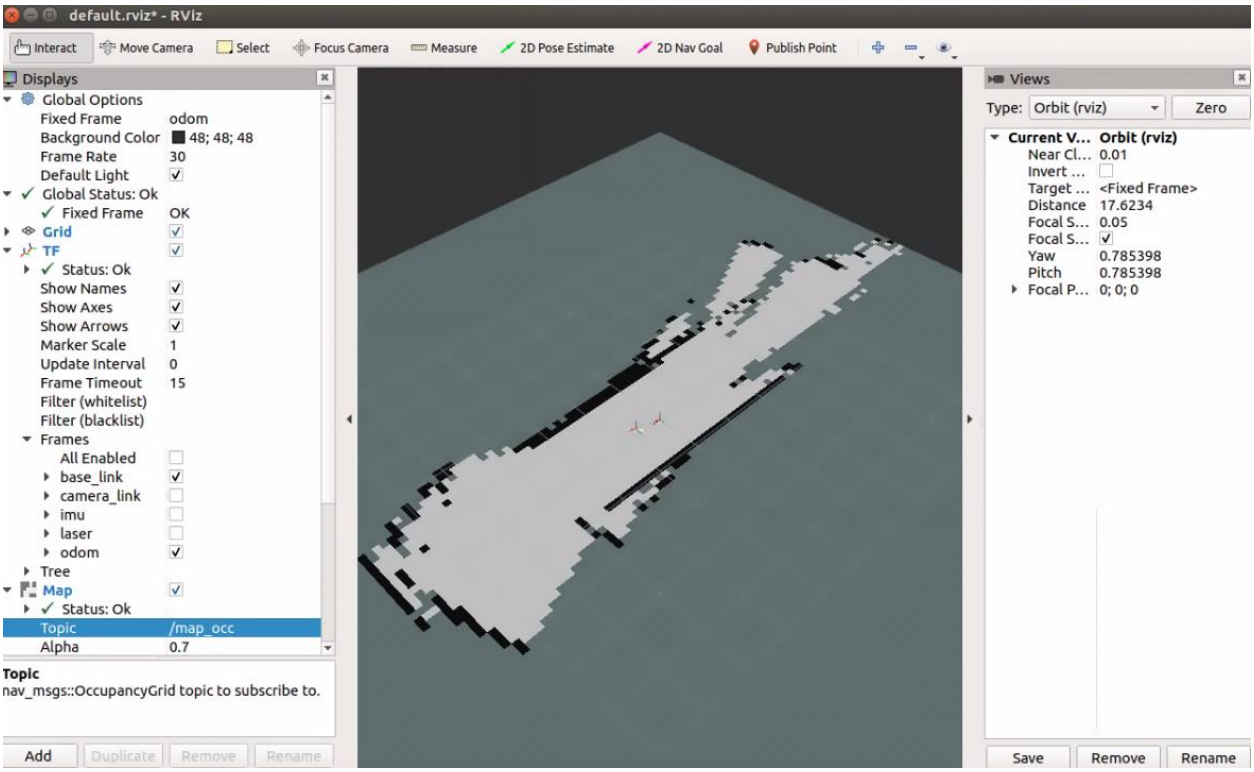
Screenshots of the RVIZ environment for manual testing:



This was our original result. Unable to detect anything recognizable.



Once we increased our p_{occ} from 0.75 to 0.9 and decreased p_{free} from 0.25 to 0.1, the algorithm seems to detect any walls/obstructions better, but still not our desired outcome.



After adjusting our map_res from 0.1 to 0.15, we started receiving more consistent results as our cell size is now larger. Not as desirable as we wanted but a cleaner result than others. We would like to note that passing students in the hallway would have affected our results. With these changes in place, the algorithm became more accurate, but at a cost of a slower program due to stricter conditions of p_{occ} and p_{free} which causes the program to take longer determining whether a cell is occupied or free.

Our final parameters:

```
# Occupancy grid map parameters
occ_map_topic: "map_occ"
p_occ: 0.9
p_free: 0.1
map_width: 120
map_height: 120
map_res: 0.15 #m
object_size: .05 #m
```

Impact of map_res

Increasing the map resolution from 0.1 m to 0.15 m effectively increases the physical area covered by each individual cell. This results in a "cleaner" map because the larger cell size acts as a spatial filter, averaging out small LiDAR measurement errors or slight vehicle pose jitters that might have caused noise

at a finer resolution. While this provides more consistent results, the trade-off is a loss of detail; obstacles smaller than 0.15m may be merged or misrepresented, and the boundaries of walls will appear blockier.

Impact of p_{occ} and p_{free}

The p_{occ} and p_{free} parameters define the confidence thresholds required to classify a cell's status. By moving these values further from the "unknown" state of 0.5 (making them "stricter"), it increased the algorithm's robustness against transient sensor noise. Since the threshold is higher, the recursive log-odds update requires more consistent LiDAR returns on the same cell before the probability crosses the classification boundary. This prevents "flickering" obstacles caused by temporary LiDAR interference. A stricter threshold means that cells with only a few LiDAR hits will remain in the "unknown" state (p=0.5 or -1 in the message) for a longer period.

A primary limitation of the implemented Occupancy Grid Mapping algorithm is the static environment assumption, which poses that the state of any given cell does not change over time. This creates significant challenges when attempting to detect or account for moving obstacles. Because the algorithm utilizes a recursive log-odds update, each cell's current classification is heavily influenced by its entire measurement history. If a cell has been consistently observed as "free" for an extended period, the accumulated negative log-odds value becomes quite large. Consequently, if a moving object suddenly enters that space, the algorithm cannot immediately reclassify the cell as "occupied." The new "occupied" signals from the LiDAR must first offset the substantial "free" history before the cell's probability can cross the occupancy threshold again. This results in the map failing to reflect the current physical state of the environment in real-time. In high-dynamic scenarios, this latency effectively renders the vehicle blind to transient obstacles, as an object would need to remain stationary for a prolonged duration to be registered on the map.

Activity 10:

In this lab, we explored the fundamental concepts of robot localization and environment mapping by implementing and testing various algorithms on the AEV. We started by finding the homogenous transformations between the base_link frame and the sensor frames which include the LiDAR, IMU, and camera. We then published them as static ROS transformations and improved our wheel odometry by replacing the yaw angle with the on in the output from the BNO055 IMU. We used a pre-recorded rosbag to ensure the code was working as expected, then combined LiDAR scan data as well as the vehicle's estimated pose to develop an Occupancy Grid Mapping node in ROS. Finally, we ensured it was working by driving the AEV in the hallway during experiments.

Some limitations of the localization and mapping algorithms is that the localization system uses the linear velocity to estimate x and y position, which means that errors such as the slight wheel slipping could accumulate over time as drift. A mapping limitation is that the IMU could be affected by magnetic interference from nearby electronics and metal. This would inaccurately report the vehicle position, which would cause the LiDAR measurements to be shown on the wrong grid cells. A limitation of the environment was that it does not account for moving obstacles such as passing pedestrians, which can leave behind occupied cells.

To address some of these limitations, we could boost the position accuracy of the AEV by implementing sensor fusion which combines the various sensor frames of odometry, IMU, and LiDAR data. For the mapping issues we could add code that allows cells to revert to unknown status when no recent LiDAR returns show that there is an object in the grid cell, which would make it more adaptable to moving obstacles.